

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1557

Lime Creek 138 kV
November 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1557. The Interconnection Customer proposes a 140.2 MW solar photovoltaic facility interconnecting at the Lime Creek 138 kV substation in the Pennsylvania area. The request was received on August 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 535 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,266,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Walmart-Keswick Substation 138 kV
Contingency: Pacific Way-Crystal Falls 138 kV
Loading: 111.7% (Rating: 189 MVA)
DFAX: 22.5%
Pre-project loading: 95.0%
2. Facility: Fobing Road-Whillock Substation 138 kV
Contingency: Crystal Falls-Pacific Way 138 kV
Loading: 107.3% (Rating: 250 MVA)
DFAX: 22.5%
Pre-project loading: 94.8%
3. Facility: Arsenal Hill-Forbing Road 138 kV
Contingency: Crystal Falls-Pacific Way 138 kV Line
Loading: 106.8% (Rating: 234 MVA)
DFAX: 22.5%
Pre-project loading: 93.3%
4. Facility: Keswick-Arsenal Hill 138 kV
Contingency: Pacific Way Substation-Crystal Falls 138 kV line
Loading: 106.2% (Rating: 220 MVA)
DFAX: 22.5%
Pre-project loading: 91.9%
5. Facility: Lewisville-Oakland 138 kV line
Contingency: Medora-P.T.S 345 kV
Loading: 102.8% (Rating: 331 MVA)
DFAX: 7.8%
Pre-project loading: 99.5%

6. Facility: Oakland-Walter L Anderson Substation 138 kV
 Contingency: Meora-P.T.S 345 kV Line
 Loading: 102.5% (Rating: 358 MVA)
 DFAX: 7.8%
 Pre-project loading: 99.4%
7. Facility: Sierra Brooks-Walter L Anderson 138 kV L
 Contingency: Medora-P.T.S 345 kV
 Loading: 102.0% (Rating: 403 MVA)
 DFAX: 7.8%
 Pre-project loading: 99.2%
8. Facility: Sierra Brooks-Conrath Dpc Solar 138 kV
 Contingency: Medora-P.T.S 345 kV
 Loading: 101.7% (Rating: 425 MVA)
 DFAX: 7.8%
 Pre-project loading: 99.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Keswick-Walmart Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$2,058,000	48% (\$990,000)
Fobing Road-Whillock Substation 138 kV	Reconductor line	\$963,000	48% (\$463,000)
Arsenal Hill-Forbing Road 138 kV	Reconductor line	\$463,000	48% (\$223,000)
Keswick-Arsenal Hill 138 kV	Reconductor line	\$588,000	48% (\$283,000)
Lewisville-Oakland 138 kV line	Reconductor line	\$699,000	9% (\$60,000)
Oakland-Walter LAnderson 138 kV L	Reconductor line	\$1,000,000	9% (\$86,000)
Sierra Brooks-Walter L Anderson 138 kV L	Reconductor line	\$1,623,000	9% (\$140,000)
Sierra Brooks-Conrath Dpc Solar 138 kV	Reconductor line	\$247,000	9% (\$21,000)
	Total allocated to project		\$2,266,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.